

ACID-BASE DISTURBANCES

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OUTLINE

- Introduction
- Acid-base homeostasis
- Basic pathophysiology
- Metabolic acidosis
- Metabolic alkalosis
- Respiratory acidosis
- Respiratory Alkalosis

Introduction

Acid-Base chemistry

- An acid is a substance that can *donate* protons $[H^+]$:
(Acid) $HCl \rightarrow H^+ + \text{Chloride ion (Cl)}$
- A base is a substance that can *accept* protons $[H^+]$:
Ammonia $(NH_3) + H^+ \rightarrow NH_4^+$ (base)
- The **acidity of body** fluids is quantified in terms of the hydrogen ion concentration.
 - By convention, the degree of acidity is expressed as pH

Introduction...

- Normally, the pH of blood is maintained at **7.40** ($[\text{H}^+]$ of $4 \times 10^{-8} \text{ M}$) with a range of 7.35 to 7.45.
- A condition incompatible with life can occur when:
 - $\text{pH} < 6.7$, representing a **fivefold** increase in hydrogen ion concentration.
 - $\text{pH} > 7.7$, representing a **50% decrease** in hydrogen ion concentration, is considered in.
- The hydrogen ion concentration in blood may not be indicative of that in other body compartments.

Buffers

- *Buffering*: the ability of a weak acid and its corresponding anion (base) to resist change in the pH of a solution with the addition of a strong acid or base.
- The principal extracellular buffer is the carbonic acid/bicarbonate ($\text{H}_2\text{CO}_3/\text{HCO}_3^-$) system.
- Other physiologic buffers include plasma proteins, hemoglobin, and phosphates.

Basic Pathophysiology

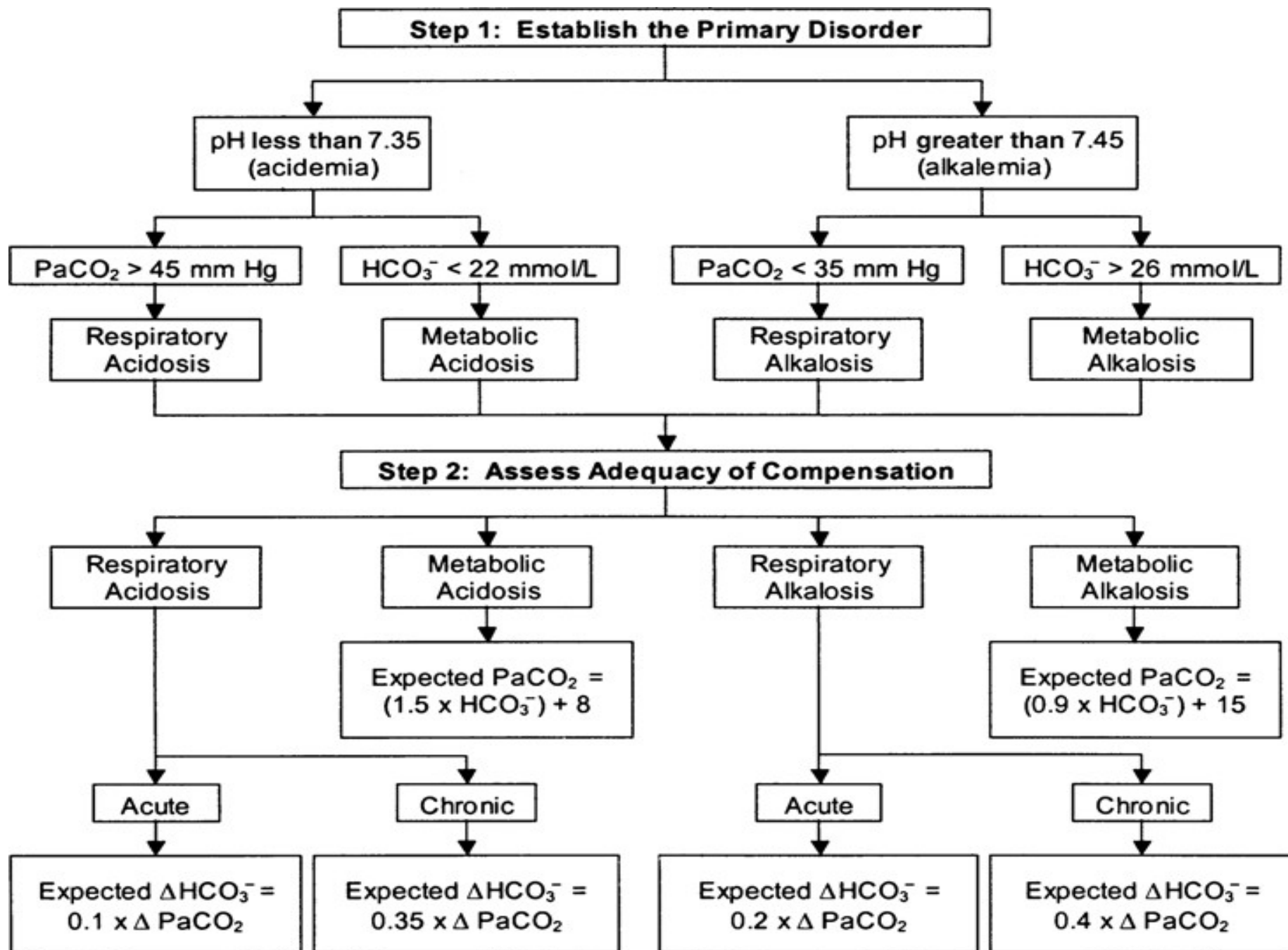
- A respiratory acid-base disorder is a pH disturbance caused by **pathologic alterations of the respiratory system** or its central nervous system control.
 - **Respiratory acidosis**
 - **Respiratory alkalosis**
- Variations in respiratory rate and/or depth allow the lungs to achieve changes in the PaCO₂ very quickly (within minutes).
- **Bicarbonate** is a **base** that is regulated by renal metabolism via the enzyme **carbonic anhydrase**--metabolic control
- **A metabolic** acid-base disorder is a PH disturbance caused by **derangement of the pathways** responsible for **maintaining a normal HCO₃ – level**.
 - Metabolic acidosis
 - Metabolic alkalosis

Acid-base homeostasis

- Acid-base homeostasis is responsible for maintaining blood hydrogen ion concentration $[H^+]$ near normal despite the daily acidic and/or alkaline loads derived from the intake and metabolism of foods.
- Blood pH is normally tightly regulated by three distinct mechanisms:
 1. Extracellular bicarbonate and intracellular protein buffering systems
 2. Pulmonary regulation of $PaCO_2$, effectively allowing carbonic acid to be eliminated by the lungs as CO_2
 3. Renal reclamation or excretion of HCO_3^- and excretion of acids such as ammonium.

Basic Pathophysiology...

- Metabolic acid-base disorders ----respiratory compensation.
- Chronic respiratory acid-base disorders--- metabolic compensation.
- Acute respiratory derangements are essentially uncompensated.



Step 3: Calculate the Anion Gap

$$\text{Anion Gap} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$$

Anion Gap < 12
No Anion Gap Metabolic Acidosis

Anion Gap 12 or more
Anion Gap Metabolic Acidosis

Step 4: Calculate the Excess Gap

$$\text{Excess Gap} = \text{Calculated Anion Gap} - 12$$

Measured HCO_3^- + Excess Gap < 22
Concurrent Non-Anion Gap Metabolic Acidosis

Measured HCO_3^- + Excess Gap > 26
Concurrent Metabolic Alkalosis

Terms and Definitions

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	$\leftarrow 7.35 - 7.45 \rightarrow$	Alkalemia
pCO ₂	Respiratory alkalosis	$\leftarrow 35 - 45 \rightarrow$	Respiratory acidosis
HCO ₃	Metabolic acidosis	$\leftarrow 22 - 26 \rightarrow$	Metabolic Alkalosis

- Respiratory compensation for metabolic disorders is rapid
- Full metabolic compensation for respiratory disturbances requires renal adjustment and takes 3-5 days

Simple-Acid Base Disorders

- Look at the pH in order to determine the primary abnormality
- Pathophysiologic principle: body does not fully compensate even for chronic acid-base disorders

Serum anion gap

- To maintain electroneutrality, the total concentration of cations in the serum must equal the total concentration of anions.

$$[\text{Na}^+] + [\text{UCs}] = ([\text{Cl}^-] + [\text{HCO}_3^-]) + [\text{UAs}] \quad \text{SAG} = [\text{Na}^+] - [\text{Cl}^-] - [\text{HCO}_3^-] \quad \text{SAG} = [\text{UAs}] - [\text{UCs}]$$

- The SAG can also be elevated in **the metabolic acidosis & helps in**
 - ❖ Identification of etiology of acid base disorders
 - ❖ Facilitate determination of excess gap or the degree to which the calculated anion exceeds the normal anion gap.
- The excess gap represents the amount of **HCO₃⁻ that has been lost** due to buffering **unmeasured cations**. The excess gap can be added back to the measured HCO₃ – to determine what the patient's bicarbonate would be if these endogenous acids were not present.

TABLE 55-2**Interpretation of Simple Acid–Base Disorders**

Acid–Base Disorder	pH	Primary Disturbances	Compensation
Acidosis			
Respiratory	Decrease	Increase P_{aCO_2}	Increase HCO_3^-
Metabolic	Decrease	Decrease HCO_3^-	Decrease P_{aCO_2}
Alkalosis			
Respiratory	Increase	Decrease P_{aCO_2}	Decrease HCO_3^-
Metabolic	Increase	Increase HCO_3^-	Increase P_{aCO_2}

Example #2

pH	7.25
pCO₂	60
HCO₃	26

Acute Respiratory Acidosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

Respiratory Acidosis

- Respiratory acidosis is characterized by :
 1. a reduced arterial pH
 2. a primary increase in the arterial PaCO_2 and, when present for sufficient time
 3. a compensatory rise in the HCO_3^- concentration.
- Because increased CO_2 is a potent respiratory stimulus, respiratory acidosis represents **ventilatory failure** or **impaired central control of ventilation** as opposed to an increase in CO_2 production.
- As such, most patients will have hypoxemia in addition to hypercapnia.

Clinical presentation

- Severe, acute respiratory acidosis produces a variety of **neurologic abnormalities**.
- Initially these include headache, blurred vision, restlessness, and anxiety.
- These may progress to tremors, asterixis, somnolence, and/or delirium.
- If untreated, terminal manifestations include **peripheral vasodilation** leading to **hypotension** and **cardiac arrhythmias**.
- Chronic respiratory acidosis is typically associated with **cor pulmonale** and **peripheral edema**.

Causes of Acute Respiratory Acidosis

- CNS depression—drugs, CNS event
- Neuromuscular disorders—myopathies, neuropathies
- Acute airway obstruction—upper airway, laryngospasm, bronchospasm
- Severe pneumonia or pulmonary edema
- Impaired lung motion—hemothorax, pneumothorax
- Thoracic cage injury—flail chest
- Ventilator dysfunction

Treatment

- In order to effectively treat respiratory acidosis, the causative process must be identified and treated.
- If a cause is identified, specific therapy should be started
 - This may include **naloxone** for opiate-induced hypoventilation or **bronchodilator** therapy for acute bronchospasm
 - Because respiratory acidosis represents ventilatory failure, an increase in alveolar ventilation is required
 - Sodium bicarbonate????
 - Excessive oxygen???

Example #1

pH	7.50
pCO ₂	29
HCO ₃	22

Acute Respiratory Alkalosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

Respiratory Alkalosis

- Respiratory alkalosis is characterized by :
 1. an increased arterial pH
 2. a primary decrease in the arterial PaCO_2 and, when present for sufficient time
 3. a compensatory fall in the HCO_3^- –concentration.
- Respiratory alkalosis represents hyperventilation and is remarkably common.

Causes of Acute Respiratory Alkalosis

- Anxiety
- Hypoxia
- Lung disease with or without hypoxia
- CNS disease
- Drug use—salicylates, catecholamines, progesterone
- Pregnancy
- Sepsis
- Hepatic Encephalopathy
- Mechanical Ventilation

Clinical presentation

- The symptoms produced by respiratory alkalosis result from increased irritability of the central and peripheral nervous systems.
- These include light-headedness, altered consciousness, distal extremity paresthesias, circumoral paresthesia, cramps, carpopedal spasms, and syncope.
- Various supraventricular and ventricular cardiac arrhythmias may occur in extreme cases, particularly in critically ill patients.
- Hypophosphatemia , reflecting a shift of phosphate from the extracellular space into the cells.

TREATMENT

- Respiratory alkalosis is usually not a severe disorder; treatment involves correcting the underlying causes.
- In case of acute anxiety the patient should rebreath **expired air** from a paper bag.
- In order to effectively treat metabolic alkalosis, the *causative process must be identified and treated*.
- Always look for administration of compounds such as **citrate** in blood products and **acetate** in parenteral nutrition that can raise the HCO_3^- level.
- If the etiology of the metabolic alkalosis is still unclear, measurement of the urinary chloride may be useful.
- ✓ Some processes leading to metabolic alkalosis (vomiting, nasogastric suction losses, and factitious diarrhea) will have low urinary Cl^- levels (less than 25 mEq/L or mmol/L)
- ✓ While others (diuretics, hypokalemia, and mineralocorticoid excess) will have higher urinary Cl^- levels (greater than 40 mEq/L or mmol/L).

Example #4

pH	7.50
pCO ₂	48
HCO ₃	36

Metabolic Alkalosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

METABOLIC ACIDOSIS

- is characterized by a decrease in pH as the result of **a primary decrease in serum bicarbonate concentration.**
- This can result from:
 - ✗ The buffering (consumption of HCO_3^-) of exogenous acid
 - ✗ An organic acid accumulating because of a metabolic disturbance (e.g., lactic acid or ketoacids), or
 - ✗ The progressive accumulation of **endogenous acids** secondary to impaired renal function (**e.g., phosphates and sulfates**).
- The serum HCO_3^- can also be decreased as the result of;
 - ✗ A loss of bicarbonate-rich body fluids (e.g., diarrhea, biliary drainage, or pancreatic fistula) or
 - ✗ Occur secondary to the rapid administration of non-alkali-containing intravenous fluids (dilutional acidosis).

Causes of Metabolic Acidosis

- Non-Gap

- GI HCO_3 loss: diarrhea, ureteral diversions
- Renal HCO_3 loss: RTA, aldosterone inhibitors, carbonic anhydrase inhibitors
- Iatrogenic: normal saline

- Anion Gap

- Ketoacidosis: diabetic, alcoholic
- Renal failure
- Lactic Acidosis
- Rhabdomyolysis
- Toxins: methanol, ethylene glycol, paraldehyde, salicylates

Example #5

pH	7.20
pCO ₂	21
HCO ₃	8

Metabolic Acidosis with Respiratory Compensation

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

Treatment

- In order to effectively treat metabolic acidosis, the *causative process must be identified and treated*.
- The role of adjunctive therapy with sodium bicarbonate (NaHCO_3) is not universally agreed upon.
- The metabolic acidosis seen with lactic acidosis and ketoacidosis generally resolves with **therapy targeted at the underlying cause** and NaHCO_3 may be unnecessary regardless of the pH.
- The metabolic acidosis of renal failure, renal tubular acidosis, or intoxication with ethylene glycol, methanol, or salicylates is much more likely to require **NaHCO_3 therapy**.

Metabolic alkalosis

- Metabolic alkalosis is characterized by:
 1. an increased arterial pH
 2. a primary increase in the HCO_3^- concentration, and
 3. compensatory increase in the PaCO_2 .
- **Common Causes of Metabolic Alkalosis**

Urine Cl^- less than 10 mEq/L (less than 10 mmol/L)	Urine Cl^- greater than 10 mEq/L (greater than 10 mmol/L)
Alkali administration - IV bicarbonate therapy - Oral alkali therapy - Parenteral nutrition with acetate "Contraction alkalosis" - post-diuretic use Decreased chloride intake - Loss of gastric acid - Vomiting - Nasogastric suction	Drugs ^a - Corticosteroid therapy - Diuretics Hypokalemia - Mineralocorticoid excess - Hyperaldosteronism - Bartter's syndrome - Cushing's syndrome

Causes of Metabolic Alkalosis

- Vomiting
- Diuretics
- Excess mineralocorticoid activity—Cushing's syndrome, Conn's syndrome, exogenous steroids, licorice ingestion, increased renin states, Bartter's syndrome
- Excess alkali administration
- Refeeding alkalosis

Diagnosis

- Healty history
- Medication history
- Aterial pH >7.45
- Bicarbonate levels >26 mmol/L

TREATMENT

- In order to effectively treat metabolic alkalosis, the *causative process must be identified and treated*.
- Always look for administration of compounds such as **citrate** in blood products and **acetate** in parenteral nutrition that can raise the HCO_3^- level.
- If the etiology of the metabolic alkalosis is still unclear, measurement of the urinary chloride may be useful.
- ✓ Some processes leading to metabolic alkalosis (vomiting, nasogastric suction losses, and factitious diarrhea) will have low urinary Cl^- levels (less than 25 mEq/L or mmol/L)
- ✓ While others (diuretics, hypokalemia, and mineralocorticoid excess) will have higher urinary Cl^- levels (greater than 40 mEq/L or mmol/L).

Mixed Acid-Base Disorders

The Rules:

- Look at the pH: whichever side of 7.40 the pH is on, the process that caused it to shift to that side is the primary abnormality
 - Principle: the body doesn't fully compensate for primary acid-base disorders
- Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$: if the anion gap is >20 , there is a primary metabolic acidosis regardless of pH or HCO_3
 - Principle: the body doesn't generate a large anion gap to compensate for a primary disorder
- Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO_3 concentration, if >30 , there is underlying metabolic alkalosis; if <24 , there is underlying non-gap metabolic acidosis
 - Principle: 1 mmol of unmeasured acid titrates 1 mmol of bicarbonate

Example #8

pH	7.50
pCO ₂	20
HCO ₃	15
Na	145
Cl	100

**Respiratory alkalosis,
Anion Gap Metabolic
Acidosis and Metabolic
Alkalosis**

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

1. Look at the pH to determine the primary process.
2. Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$
3. Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO₃ concentration, if >30, there is underlying metabolic alkalosis; if <24, there is underlying non-gap metabolic acidosis

Example #6

pH	7.50
pCO ₂	20
HCO ₃	15
Na	140
Cl	103

Respiratory Alkalosis and Anion Gap Metabolic Acidosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

1. Look at the pH to determine the primary process.
2. Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$
3. Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO₃ concentration, if >30, there is underlying metabolic alkalosis; if <24, there is underlying non-gap metabolic acidosis

Example #7

pH	7.40
pCO ₂	40
HCO ₃	24
Na	145
Cl	100

Anion Gap Metabolic Acidosis and Metabolic Alkalosis

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

1. Look at the pH to determine the primary process.
2. Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$
3. Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO₃ concentration, if >30, there is underlying metabolic alkalosis; if <24, there is underlying non-gap metabolic acidosis

Example #9

pH	7.10
pCO ₂	50
HCO ₃	15
Na	145
Cl	100

**Respiratory Acidosis,
Anion gap Metabolic
Acidosis, Metabolic
Alkalosis**

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

1. Look at the pH to determine the primary process.
2. Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$
3. Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO₃ concentration, if >30, there is underlying metabolic alkalosis; if <24, there is underlying non-gap metabolic acidosis

Example #10

pH	7.15
pCO ₂	15
HCO ₃	5
Na	140
Cl	110

Anion Gap and Non-Anion Gap Metabolic Acidoses

Variable	Primary Disorder	Normal Range, arterial Gas	Primary Disorder
pH	Acidemia	←7.35-7.45→	Alkalemia
pCO ₂	Respiratory alkalosis	←35 - 45→	Respiratory acidosis
HCO ₃	Metabolic acidosis	←22 – 26→	Metabolic Alkalosis

1. Look at the pH to determine the primary process.
2. Calculate the anion gap: $\text{Na} - (\text{Cl} + \text{HCO}_3)$
3. Calculate the excess anion gap (total anion gap minus the normal anion gap) and add this to the measured HCO₃ concentration, if >30, there is underlying metabolic alkalosis; if <24, there is underlying non-gap metabolic acidosis

Conclusions

- Acid-base disturbances are easy to analyze if approached systematically
- Determine primary abnormalities based on pH
- Calculate the anion gap
- Calculate the delta gap and add to the measured HCO_3
- Calculate an anion gap on EVERY chemistry you see
- If there is an elevated anion gap, remember to get an ABG!!